SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Cryptography and Network Security
COURSE CODE: CSA51

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 20%

QUESTIONS: 5

Total Marks:50

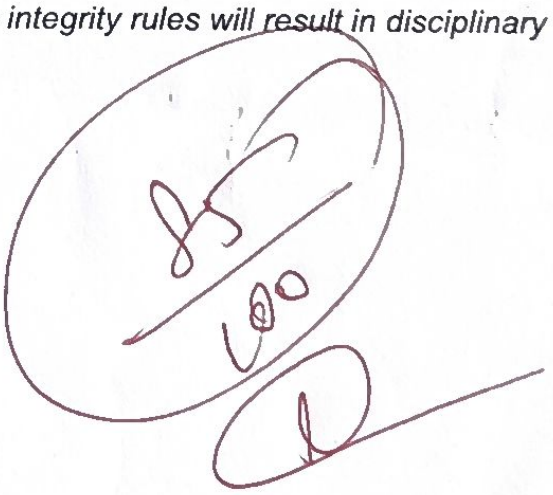
Annexure A

CONCEPT MAPPING

Code of Conduct:

I C. Priyanka/192511014 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Annexure A

CONCEPT MAPPING

1. Create a concept map linking the following: Security Attacks, Threats, Risks and Security Goals (Confidentiality, Integrity, Availability. Perform the following tasks:
    a. Show how each attack leads to specific threats and risks
    b. Map how security goals help mitigate them Provide one real-world example for each link

2. Develop a concept map comparing substitution techniques: Caesar Cipher, Monoalphabetic Cipher, Vigenère Cipher and Hill Cipher. Perform the following tasks to connect based on: Key usage, Complexity and Security strength. Analyze which technique is most secure and why.

3. Construct a concept map for Classical Encryption Schemes: Substitution Techniques and Transposition Techniques. Perform the following tasks:
    a. Show differences and similarities
        i. Map how each affects the Data structure and Security level
        ii. Analyze which is more vulnerable to attacks and why

4. Create a concept map connecting Steganography concepts: Data Hiding, Watermarking and Survivability. Perform the following tasks:
    a. Show relationships between these concepts
    b. Analyze how each contributes to secure communication
    c. Include one real-world application for each

5. Draw a concept map linking number theory concepts used in cryptography: Modular Inverse, Extended Euclid Algorithm, Fermat's Little Theorem, Euler's Phi Function and Euler's Theorem. Perform the following tasks:
    a. Show how each concept is mathematically connected
    b. Map their role in encryption/decryption
    c. Analyze which concept is most critical for public key cryptography and justify

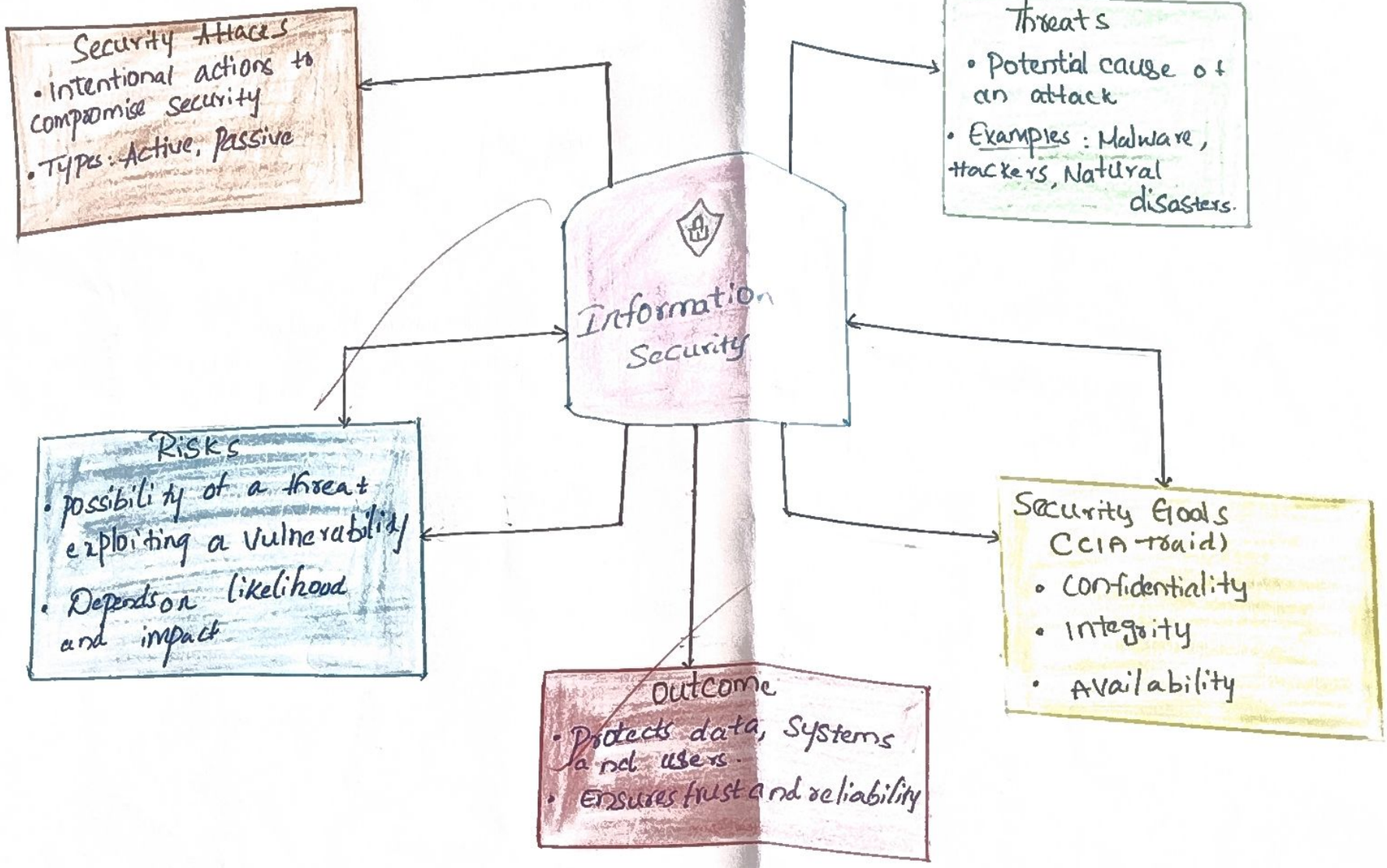
RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

25 20
30 27
20 20
15 13
10 10

85/100

Security Attacks, Threats, Risks and Security Goals



Number theory concepts in Cryptography

Number theory concepts
Mathematically-foundation for modern cryptography

Modular Inverse
- Multiplicative inverse in modular arithmetic
- Used in RSA (decryption key)
- Solved using Extended Euclid Algorithm
- a^{-1} mod m, such that $(a \times a^{-1}) \equiv 1 \pmod{m}$

Extended Euclid Algorithm
- Finds GCD and modular inverse
- Solves linear Diophantive equations.
- Used to compute private key in RSA

Euler's Theorem
- Generalization of Fermat's little theorem
- If gcd $(a, n) = 1$, then $a^{\phi(n)} \equiv 1 \pmod{n}$
- Used in public key Cryptography.

Fermat's Little Theorem
- If p is a prime, $a^{(p-1)} \equiv 1 \pmod{p}$
- Used in primality testing
- Forms the basis for RSA encryption/decryption

Euler's phi Function
- $\phi(n)$ = number of integers $(1 \le k \le n)$ that are coprime to n
- Used in RSA to compute private key.
- For prime p, $\phi(p) = p - 1$

Mathematical Connection
- Fermat's little Theorem is a special case of Euler's theorem (when n is prime)
- Euler's phi function is used in Euler's theorem.
- Extended Euclid Algorithm helps in finding Modular inverse, which is required in RSA.